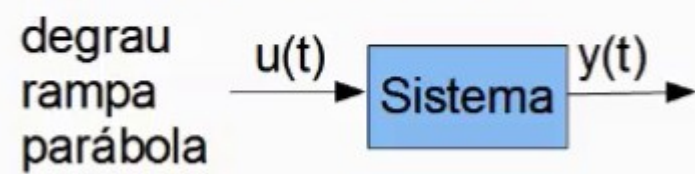
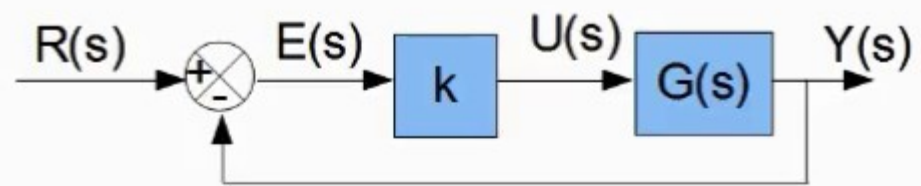


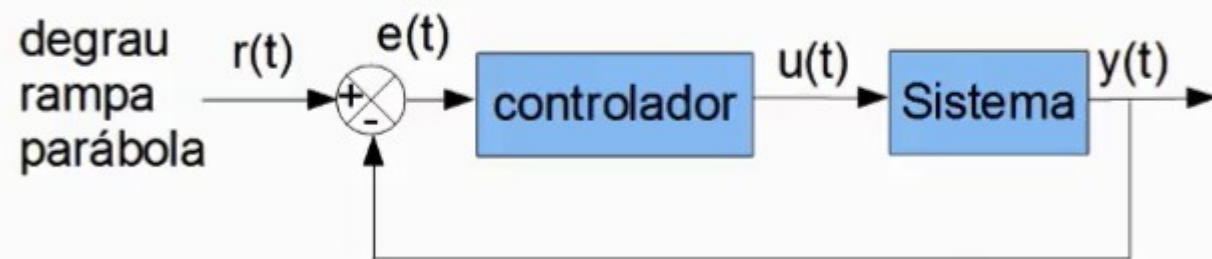
Malha Aberta:

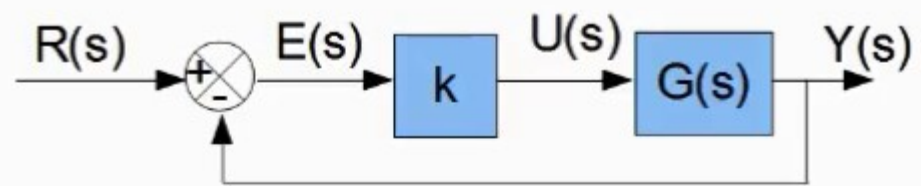




$$T(s) = \frac{k G(s)}{1 + k G(s)}$$

Malha Fechada:





$$T(s) = \frac{k G(s)}{1 + k G(s)}$$

$$T(s) = \frac{k G(s)}{1 + k G(s)}$$

$$L(s) = k G(s)$$

$$T(s) = \frac{L(s)}{1 + L(s)}$$

$$E(s) = R(s) - Y(s) = R(s) - T(s)R(s)$$

$$E(s) = R(s) - \frac{L(s)}{1 + L(s)} R(s)$$

$$E(s) = R(s) - Y(s) = R(s) - T(s)R(s)$$

$$E(s) = R(s) - \frac{L(s)}{1 + L(s)} R(s)$$

$$E(s) = R(s) \frac{1}{1 + L(s)}$$

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Valor final $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s \frac{1}{s} \frac{1}{1 + L(s)} = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}$$

$$e_{\infty} = \frac{1}{1 + \lim_{s \rightarrow 0} L(s)}$$

$$k_p = \lim_{s \rightarrow 0} L(s)$$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s \frac{1}{s} \frac{1}{1 + L(s)} = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}$$

$$e_{\infty} = \frac{1}{1 + \lim_{s \rightarrow 0} L(s)} \quad e_{\infty(\text{degrau})} = \frac{1}{1 + k_p}$$

$$k_p = \lim_{s \rightarrow 0} L(s)$$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1 + L(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sL(s)}$$

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$$e_{\infty} = \frac{1}{\lim_{s \rightarrow 0} sL(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1 + L(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sL(s)}$$

$$e_{\infty} = \frac{1}{\lim_{s \rightarrow 0} sL(s)} \qquad e_{\infty(\text{rampa})} = \frac{1}{k_v}$$

$$k_v = \lim_{s \rightarrow 0} sL(s)$$

$$e_{\infty} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} R(s) \frac{1}{1 + L(s)}$$

$$e_{\infty} = \lim_{s \rightarrow 0} s \frac{1}{s^3} \frac{1}{1 + L(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 L(s)}$$

$$e_{\infty} = \frac{1}{\lim_{s \rightarrow 0} s^2 L(s)} \qquad e_{\infty(\text{parábola})} = \frac{1}{k_a}$$

$$k_a = \lim_{s \rightarrow 0} s^2 L(s)$$

$$e_{\infty(\text{degrau})} = \frac{1}{1 + k_p}$$

$$e_{\infty(\text{rampa})} = \frac{1}{k_v}$$

$$e_{\infty(\text{parábola})} = \frac{1}{k_a}$$

$$k_p = \lim_{s \rightarrow 0} L(s)$$

$$k_v = \lim_{s \rightarrow 0} sL(s)$$

$$k_a = \lim_{s \rightarrow 0} s^2 L(s)$$

$$L(s) = k \cdot G(s) = k \frac{N_G(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

$$L(s) = k G(s) = k \frac{N_G(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

$$k_p = k \frac{N_G(0)}{(-p_1)(-p_2) \cdots (-p_n)} = k \frac{N_G(0)}{N_D(0)} = k G(0)$$

$$k_v = k \lim_{s \rightarrow 0} s \frac{N_G(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)} = k G(0) \lim_{s \rightarrow 0} s = 0$$

$$k_a = k \lim_{s \rightarrow 0} s^2 \frac{N_G(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)} = k G(0) \lim_{s \rightarrow 0} s^2 = 0$$

$$L(s) = k G(s) = k \frac{N_G(s)}{s(s - p_2) \cdots (s - p_n)}$$

$$k_p = k \frac{1}{\lim_{s \rightarrow 0} s} \frac{N_G(0)}{(-p_2) \cdots (-p_n)} \rightarrow \infty$$

$$k_v = k \lim_{s \rightarrow 0} \frac{N_G(s)}{(s - p_2) \cdots (s - p_n)} = k \frac{N_G(0)}{(-p_2) \cdots (-p_n)}$$

$$k_a = k \lim_{s \rightarrow 0} s \frac{N_G(s)}{(s - p_2) \cdots (s - p_n)} = k \lim_{s \rightarrow 0} s G(s) \lim_{s \rightarrow 0} s = 0$$

$$L(s) = k G(s) = k \frac{N_G(s)}{s^2 (s - p_3) \cdots (s - p_n)}$$

$$k_p = k \lim_{s \rightarrow 0} \frac{1}{s^2} \frac{N_G(0)}{(-p_3) \cdots (-p_n)} \rightarrow \infty$$

$$k_v = k \lim_{s \rightarrow 0} \frac{1}{s} \frac{N_G(0)}{(-p_3) \cdots (-p_n)} \rightarrow \infty$$

$$k_a = k \lim_{s \rightarrow 0} \frac{N_G(s)}{(s - p_3) \cdots (s - p_n)} = k \frac{N_G(0)}{(-p_3) \cdots (-p_n)}$$

Quantidade de polos na origem

	0	1	2
k_p	$k G(0)$	∞	∞
k_v	0	$k \lim_{s \rightarrow 0} s G(s)$	∞
k_a	0	0	$k \lim_{s \rightarrow 0} s^2 G(s)$

Tipo numérico do Sistema.

	Tipo 0	Tipo 1	Tipo 2
k_p	$k G(0)$	∞	∞
k_v	0	$k \lim_{s \rightarrow 0} s G(s)$	∞
k_a	0	0	$k \lim_{s \rightarrow 0} s^2 G(s)$

Tipo numérico do Sistema.

	Tipo 0	Tipo 1	Tipo 2
$e_{\infty}(d)$	$\frac{1}{1+k G(0)}$	0	0
$e_{\infty}(r)$	∞	$\frac{1}{k \lim_{s \rightarrow 0} s G(s)}$	0
$e_{\infty}(p)$	∞	∞	$\frac{1}{k \lim_{s \rightarrow 0} s^2 G(s)}$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s + a_0}$$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s + a_0}$$

$$G(0) = \frac{b_0}{a_0} \quad k_p = k \frac{b_0}{a_0}$$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s}$$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s(s^{n-1} + a_{n-1} s^{n-2} + a_{n-2} s^{n-3} + \dots + a_2 s + a_1)}$$

$$k_v = k \frac{b_0}{a_1}$$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_2 s^2}$$

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^2 (s^{n-2} + a_{n-1} s^{n-3} + \dots + a_3 s + a_2)}$$

$$k_a = k \frac{b_0}{a_2}$$

Tipo numérico do Sistema.

	Tipo 0	Tipo 1	Tipo 2
$e_{\infty}(d)$	$\frac{a_0}{a_0 + k b_0}$	0	0
$e_{\infty}(r)$	∞	$\frac{a_1}{k b_0}$	0
$e_{\infty}(p)$	∞	∞	$\frac{a_2}{k b_0}$